Linear Regression: Probabilistic Interpretation (MLE) and Matrix Derivation

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1. Probabilistic View of Linear Regression

[cite_start]Instead of just choosing a line blindly, we assume that the target variable y_i is generated by a deterministic linear combination of inputs plus some random, unobserved noise term ϵ_i[cite: 32]:

$$y_i = \beta^T \mathbf{x}_i + \epsilon_i$$

The Gaussian Noise Assumption

[cite_start]We assume that the error terms ϵ_i are **Independently and Identically Distributed (i.i.d.)** following a Normal (Gaussian) distribution with zero mean and constant variance σ^2[cite: 32, 41]:

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

[cite_start]The Probability Density Function (PDF) of a single noise component is given by[cite: 34, 35]:

$$p(\epsilon_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\epsilon_i^2}{2\sigma^2}\right)$$

[cite_start]Because $\epsilon_i = y_i - \beta^T \mathbf{x}_i$, it follows that given the input features $\mathbf{x}_i$ parameterized by β, the target y_i also follows a Gaussian distribution[cite: 32, 37]:

$$y_i \mid \mathbf{x}_i; \beta \sim \mathcal{N}(\beta^T \mathbf{x}_i, \sigma^2)$$

$$p(y_i \mid \mathbf{x}_i; \beta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta^T \mathbf{x}_i)^2}{2\sigma^2}\right)$$

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2. Maximum Likelihood Estimation (MLE)

[cite_start]Since the samples are independent, the joint probability of seeing our entire dataset (the **Likelihood function**, $L(\beta)$) is the product of the individual probabilities[cite: 41, 51]:

$$L(\beta) = \prod_{i=1}^{n} p(y_i \mid \mathbf{x}_i; \beta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta^T \mathbf{x}_i)^2}{2\sigma^2}\right)$$

[cite_start]Our goal is to find the parameter vector β that maximizes this likelihood[cite: 43].

The Log-Likelihood Transformation

[cite_start]Because the logarithm is a strictly increasing (monotonic) function, maximizing $L(\beta)$ is completely equivalent to maximizing the **Log-Likelihood** $\ell(\beta) = \log L(\beta)$[cite: 49, 53, 54]. [cite_start]Transforming products into sums makes optimization significantly easier[cite: 54]:

$$\ell(\beta) = \log\left(\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta^T \mathbf{x}_i)^2}{2\sigma^2}\right)\right)$$

$$\ell(\beta) = \sum_{i=1}^{n} \left[\log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{(y_i - \beta^T \mathbf{x}_i)^2}{2\sigma^2}\right]$$

$$\ell(\beta) = n \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta^T \mathbf{x}_i)^2$$

Equivalence to Ordinary Least Squares (OLS)

[cite_start]To maximize $\ell(\beta)$ with respect to β, we can ignore the constant left-hand term[cite: 57]. [cite_start]Maximizing a negative expression is exactly identical to minimizing its positive counterpart[cite: 55]:

$$\max_{\beta} \ell(\beta) \iff \min_{\beta} \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta^T \mathbf{x}_i)^2$$

[cite_start]Since σ^2 is a constant positive scaling factor, we can omit it as well[cite: 33, 57]. [cite_start]This leaves us with the exact foundational optimization definition of Least Squares[cite: 57]:

$$\min_{\beta} \frac{1}{2} \sum_{i=1}^{n} (y_i - \beta^T \mathbf{x}_i)^2$$

> **Key Takeaway:** Minimizing the squared errors is not an arbitrary choice! [cite_start]It is the mathematically rigorous consequence of assuming your data contains Gaussian-distributed noise[cite: 32, 57].

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3. Vectorized Matrix Representation

[cite_start]To handle multiple features efficiently without writing giant sigma summations, we group our data observations into vectors and matrices[cite: 59, 61].

[cite_start]Let our training set contain n instances and p features[cite: 14]:

- [cite_start]**Feature Matrix ($\mathbf{X}$)**: Size $n \times p$ [cite: 14, 59]
- [cite_start]**Target Vector ($\mathbf{y}$)**: Size $n \times 1$ [cite: 14, 61]
- [cite_start]**Parameter Vector (β)**: Size $p \times 1$ [cite: 61, 62]

$$\mathbf{X} = \begin{pmatrix} —(\mathbf{x}_1)^T— \\ —(\mathbf{x}_2)^T— \\ \vdots \\ —(\mathbf{x}_n)^T— \end{pmatrix} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix} \in \mathbb{R}^p$$

[cite_start]Using matrix multiplication, the vector of all predictions is[cite: 63]:

$$\mathbf{X}\beta = \begin{pmatrix} \mathbf{x}_1^T\beta \\ \mathbf{x}_2^T\beta \\ \vdots \\ \mathbf{x}_n^T\beta \end{pmatrix} \in \mathbb{R}^n$$

[cite_start]Thus, the global residual error vector is defined as[cite: 64]:

$$\mathbf{y} - \mathbf{X}\beta = \begin{pmatrix} y_1 - \mathbf{x}_1^T\beta \\ y_2 - \mathbf{x}_2^T\beta \\ \vdots \\ y_n - \mathbf{x}_n^T\beta \end{pmatrix}$$

[cite_start]We rewrite the global loss function $L(\beta)$ as the inner dot product of this error vector with itself[cite: 13, 65]:

$$L(\beta) = \frac{1}{2}(\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)$$

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4. Matrix Calculus Optimization (Closed-Form Solution)

Let us expand the matrix loss function using the properties of transposes: $(\mathbf{A} - \mathbf{B})^T = \mathbf{A}^T - \mathbf{B}^T$ and $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

$$L(\beta) = \frac{1}{2}\left(\mathbf{y}^T - (\mathbf{X}\beta)^T\right)(\mathbf{y} - \mathbf{X}\beta)$$

$$L(\beta) = \frac{1}{2}\left(\mathbf{y}^T - \beta^T\mathbf{X}^T\right)(\mathbf{y} - \mathbf{X}\beta)$$

[cite_start]

$$L(\beta) = \frac{1}{2}\left(\mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\beta - \beta^T\mathbf{X}^T\mathbf{y} + \beta^T\mathbf{X}^T\mathbf{X}\beta\right) \quad \text{[cite: 18]}$$

Because $\mathbf{y}^T\mathbf{X}\beta$ is a scalar (1×1 dimension value), it is identical to its own transpose:

$$(\mathbf{y}^T\mathbf{X}\beta)^T = \beta^T\mathbf{X}^T\mathbf{y}$$

[cite_start]We can combine these middle two twin scalar terms[cite: 19]:

[cite_start]

$$L(\beta) = \frac{1}{2}\left(\mathbf{y}^T\mathbf{y} - 2\beta^T\mathbf{X}^T\mathbf{y} + \beta^T\mathbf{X}^T\mathbf{X}\beta\right) \quad \text{[cite: 19]}$$

Taking the Gradient with Respect to β

To find the minimum point, we calculate the gradient $\nabla_\beta L(\beta)$ and set it to the zero vector $\mathbf{0}$. [cite_start]We use two core rules of Matrix Calculus[cite: 15]: 1. **Linear Rule:** $\frac{\partial(\mathbf{a}^T\mathbf{x})}{\partial\mathbf{x}} = \mathbf{a}$ 2. **Quadratic Form Rule:** $\frac{\partial(\mathbf{x}^T\mathbf{A}\mathbf{x})}{\partial\mathbf{x}} = (\mathbf{A}+\mathbf{A}^T)\mathbf{x}$. [cite_start]Since $\mathbf{X}^T\mathbf{X}$ is a symmetric matrix, this simplifies nicely to $2\mathbf{X}^T\mathbf{X}\mathbf{x}$[cite: 16, 17].

[cite_start]Applying these derivative rules to each component of our loss function[cite: 18, 20]:

[cite_start]

$$\nabla_\beta L(\beta) = \frac{1}{2}\left(\mathbf{0} - 2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\beta\right) = \mathbf{0} \quad \text{[cite: 20]}$$

[cite_start]

$$-\mathbf{X}^T\mathbf{y} + \mathbf{X}^T\mathbf{X}\beta = \mathbf{0} \quad \text{[cite: 20]}$$

[cite_start]

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y} \quad \text{[cite: 20]}$$

[cite_start]This expression is fundamentally known in linear algebra as the **Normal Equations**[cite: 20].

The Analytic Solution

[cite_start]Assuming the square matrix $\mathbf{X}^T\mathbf{X}$ is full-rank and invertible, we isolate β by multiplying both sides by $(\mathbf{X}^T\mathbf{X})^{-1}$ from the left[cite: 20]:

[cite_start]

$$\beta = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \quad \text{[cite: 20, 21]}$$

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5. Summary Reference Table

Formula Dimension Type	Analytical Equation Structure	Context / Definition
Probabilistic Sample Error	$y_i \sim \mathcal{N}(\beta^T\mathbf{x}_i, \sigma^2)$	[cite_start]Error follows normal curve distribution[cite: 32, 37].

Formula Dimension Type	Analytical Equation Structure	Context / Definition
Matrix Loss Form (RSS)	$L(\beta) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\beta\|_2^2$	[cite_start]Compresses scalar loops into compact linear algebraic form[cite: 13].
Normal Equations Matrix System	$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y}$	[cite_start]The foundational gradient zero intersection system[cite: 20].
Closed-Form Vector Solution	$\beta = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	[cite_start]Direct global optimization parameter calculation[cite: 20, 21].